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To: Editor-in-Chief, IEEE Transactions on Quantum Engineering (TQE)

Re: Submission of the manuscript "Hamiltonian Vision Kernel: Symmetry-Pooled Quantum Correlator Features with Hardware Validation"

Dear Editor,

We are submitting the enclosed manuscript, "Hamiltonian Vision Kernel: Symmetry-Pooled Quantum Correlator Features with Hardware Validation," for consideration as an Article in IEEE Transactions on Quantum Engineering.

The manuscript introduces the Hamiltonian Vision Kernel (HVK), a new hybrid quantum-classical architecture for image reconstruction that combines matrix-product-state patch features, a variational quantum circuit exposing a Pauli-correlator latent space, Fourier positional encoding, a learnable Hamiltonian energy diagnostic, and a classical decoder. We make no claim of quantum advantage: HVK is presented as a new, working architecture that is competitive with resource-matched classical baselines, validated on real quantum hardware, and equipped with capabilities — an entanglement-sensitive channel and an interpretable Hamiltonian diagnostic — that a classical reconstruction pipeline does not natively expose. We believe this fits TQE's scope directly: it sits at the intersection of quantum-circuit design, hybrid classical-quantum learning, and hardware-validated evaluation, and it takes deliberate care to characterize where the approach's evidence is strong and where it is exploratory rather than confirmatory.

Five results anchor the manuscript:

1. The pooled HVK2D observable map achieves D_4 group-equivariance by construction, with a measured numerical equivariance error of 9.57×10^{-17}.
2. On a resource-matched, multi-seed held-out CIFAR-10 comparison, a pre-declared Two One-Sided Tests (TOST) equivalence test at a ± 1 dB margin shows HVK2D is statistically equivalent to six of eight tested classical/ablated controls, while still beating the two weakest controls (random-VQC, strict classical random features) by a wide margin — the evidence that licenses calling HVK "competitive" rather than simply "not shown to differ."
3. On a leakage-audited diagnostic task requiring distant-patch correlations, the entangling observable channel reaches mean $R^2 = 0.974$ across five seeds, against $R^2 \leq 0.02$ for non-entangling controls – evidence that the model's entangling structure, not incidental capacity, is responsible for this capability.
4. Five images are reconstructed from observables measured on real IBM Quantum hardware, without retraining the decoder, reaching 25.90–31.52 dB PSNR – a scoped hardware feasibility pilot, reported with complete job metadata and simulator-transfer checks rather than as a general performance claim.

5. Exploratory training-dynamics diagnostics (a magnetization-style change-point statistic and an energy-to-entanglement ratio) are tracked across qubit count and MPS bond dimension in simulation and separately replayed on real hardware via gate-for-gate checkpoint reconstruction; we are explicit throughout that these are descriptive, small-sample observations, not inferential evidence of a genuine thermodynamic phase transition.

A companion supplementary study accompanies the manuscript with resource-matched held-out controls, leakage audits, statistical significance and equivalence testing, and complete hardware execution methodology (backend names, job IDs, shot counts). A standalone literature review accompanies both, surveying the broader hybrid quantum-classical vision literature with explicit mathematical formalism. Code, trained checkpoints, and raw result artifacts backing every reported number are maintained in a public repository (see `REPRODUCE.md` in the submission materials for a one-to-one table/figure-to-script map).

This manuscript is not under review at any other journal, and all authors have approved this submission.

[FILL: Suggested reviewers (2–3 names, affiliations, emails). Needs domain knowledge of who is active and appropriate in variational quantum algorithms / quantum machine learning / tensor-network methods for images – a judgment call for the authors and supervisor, not something to guess.]

We thank the editors and reviewers for their time and consideration.

Sincerely,

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